

Praneeth Tadepalli

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Career Interest

Actively seeking Summer 2026 internship opportunities in Analog, RF, and Mixed-Signal IC Design

Education

University of Minnesota Twin Cities, MN Sep 2025 – May 2027

MS in Electrical and Computer Engineering | GPA: 3.778/4.0

- **Coursework:** Analog IC Design(Weak Inversion, gm/Id methodology), VLSI Design I(Combinational and sequential Circuit Design, FinFETs, Delay Optimization), VLSI Design Lab(RTL-to-GDSII flow), Analog Ckts for Communications (LNA, Mixers, RF, 802.11g), VLSI Design II(SRAM, DRAM)

IIITDM Kancheepuram, Chennai, India

Dec 2021 – May 2025

B.Tech in Electronics and Communication Engineering | CGPA: 8.98/10.0

- **Coursework:** Analog Circuits, Control Systems, Signals and Systems, Communication Systems

Technical Skills

Programming: MATLAB, Python, Verilog, TCL

EDA Tools: LTSpice, Hspice, Cadence Virtuoso, Spectre, Synopsys DV, ICC2, Prime Time

Experience

Mobile & Cloud Computing Laboratory - University of Hyderabad, India

May 2024 - Sep 2024

Research Intern

- Designed a solar-powered charging circuit for *Raspberry Pi 4* using a solar panel, battery pack, and charge controller with voltage regulation
- Integrated safety components including capacitor, polyfuse, and switch for reliable and protected power delivery

Projects

2.4 GHz RF Front-End (LNA + 4-Path Mixer) | Cadence Virtuoso, Spectre | (Prof.

Jan 2026 - May 2026

Ramesh Harjani)

- Designed a common-source inductive degeneration LNA in *TSMC 16 nm* achieving *18 dB power gain*, *1.1 dB noise figure*, and *1 mW power* at 2.4 GHz (802.11g specs)
- Achieved simultaneous noise and input matching via source degeneration and gate matching network optimization (S_{11})
- Implemented a 4-path passive mixer with non-overlapping LO phases for low-loss frequency downconversion and harmonic rejection
- Validated performance using AC, transient, and noise simulations; evaluated gain, NF, and frequency response

Low-Power Wide Swing Cascode OTA with Switched-Capacitor Biquad Filter for

Sep 2025 - Dec 2025

ECG Signal Amplification | Cadence Virtuoso, Spectre | (Supervised by Prof.

Ramesh Harjani)

- Designed a Differential Wide Swing Cascode OTA using *TSMC 16 nm* technology, focusing on low power and compact area with a gain of *51.1 dB* and Unity Gain Frequency of *128.8 MHz*
- Validated the performance across *PVT* variations with an FOM ranging from *1175 to 1380*
- Developed a high Q switched-capacitor biquad filter for low-frequency ECG signal filtering and amplification with a *Q factor* of *1.037*

Markov Chain Monte Carlo (MCMC) Processor with Knuth–Yao Discrete Sampler

Sep 2025 - Dec 2025

Core | Synopsys DV, Prime Time, Synopsys ICC2 | (Supervised by Prof. Yang

Zhao)

- Designed a Verilog-based Knuth–Yao discrete sampler core utilizing an LFSR-based random number generator (RNG) for probabilistic sampling in MCMC processors
- Implemented and verified the RTL architecture using Synopsys Design Compiler and VCS
- Completed RTL-to-GDSII flow including synthesis, floor-planning, placement & routing, and sign-off analysis

8-bit Threshold Leaky ReLU (TLReLU) | Cadence Virtuoso, Hspice | (Supervised by Prof. Yu Cao)

Oct 2025 - Dec 2025

- Designed an 8-bit TLReLU circuit in *TSMC 16 nm* CMOS technology using a carry-select adder architecture to minimize the critical path delay
- The optimized design results in a total transistor count of *1098* and a propagation delay of *120 ps*

Analog Spiking Neural Network | Cadence Virtuoso, Spectre

Jun 2024 - May 2025

- Simulated neural circuits including *Axon Hillock*, *Voltage Amplifier I&F*, *Low-Power I&F*, and *AdEx* models in Cadence Virtuoso using *gpdk 90 nm* technology
- Implemented analog synaptic plasticity learning rules using *Pair-based* and *Triplet-based STDP* circuits with different memristor models
- Developed a *BCM-based ECG signal classifier* using Triplet STDP with a memristor circuit, achieving *98%* accuracy